

XIV.LOG ANALYSIS FOR DETECTION AND RESPONSE

Aim:

To learn about log analysis, best practices, and essential tools for effective detection and response.

Procedure:

1. Collect logs
2. Normalize (fix time/format)
3. Create timeline
4. Filter & search
5. Detect anomalies
6. Investigate
7. Respond & report

Tasks:

Task 1:Introduction

I'm ready to proceed!

No answer needed

✓ Correct Answer

Task 2: Log Analysis Basics

I understand the basics of logs and I'm ready to proceed!

No answer needed

✓ Correct Answer

Task 3: Investigation Theory

What's the term for a consolidated chronological view of logged events from diverse sources, often used in log analysis and digital forensics?

Super Timeline

✓ Correct Answer

Which threat intelligence indicator would `5b31f93c09ad1d065c0491b764d04933` and `763f8bdbbc98d105a8e82f36157e98bbe` be classified as?

File Hashes

✓ Correct Answer

Task 4: Detection Engineering

What is the default file path to view logs regarding HTTP requests on an Nginx server?

/var/log/nginx/access.log

✓ Correct Answer

A log entry containing `%2E%2E%2F%2E%2E%2Fproc%2Fself%2Fenviron` was identified. What kind of attack might this infer?

Path Traversal

✓ Correct Answer

Task 5: Automated vs Manual Analysis

A log file is processed by a tool which returns an output. What form of analysis is this?

Automated

✓ Correct Answer

An analyst opens a log file and searches for events. What form of analysis is this?

Manual

✓ Correct Answer

Task 6: Log Analysis Tools:Command Line

Use `cut` on the `apache.log` file to return only the URLs. What is the flag that is returned in one of the unique entries?

c701d43cc5a3acb9b5b04db7f1be94f6

✓ Correct Answer



In the `apache.log` file, how many total HTTP 200 responses were logged?

52

✓ Correct Answer



In the `apache.log` file, which IP address generated the most traffic?

145.76.33.201

✓ Correct Answer



What is the complete timestamp of the entry where `110.122.65.76` accessed `/login.php`?

31/Jul/2023:12:34:40 +0000

✓ Correct Answer



Task 7: Log Analysis Tools:Regular Expressions

How would you modify the original `grep` pattern above to match blog posts with an ID between 20-29?

`post=2[0-9]`

✓ Correct Answer



What is the name of the filter plugin used in Logstash to parse unstructured log data?

`Grok`

✓ Correct Answer

Task 8: Log Analysis Tools:Cyber Chef

Locate the "loganalysis.zip" file under `/root/Rooms/introloganalysis/task8` and extract the contents.

No answer needed

✓ Correct Answer

Upload the log file named "access.log" to CyberChef. Use regex to list all of the IP addresses. What is the full IP address beginning in 212?

`212.14.17.145`

✓ Correct Answer

Using the same log file from Question #2, a request was made that is encoded in base64. What is the decoded value?

`THM{CYBERCHEF_WIZARD}`

✓ Correct Answer

Using CyberChef, decode the file named "encodedflag.txt" and use regex to extract the MAC address. What is the extracted value?

`08-2E-9A-4B-7F-61`

✓ Correct Answer

Task 9: Log Analysis Tools:Yara and Sigma

What languages does Sigma use?

YAML

✓ Correct Answer

What keyword is used to denote the "title" of a Sigma rule?

title

✓ Correct Answer

What keyword is used to denote the "name" of a rule in YARA?

rule

✓ Correct Answer

Task 10: Conclusion

Click and continue learning!

No answer needed

✓ Correct Answer

Result:

Hence log analysis for detection and response was learned and implemented.